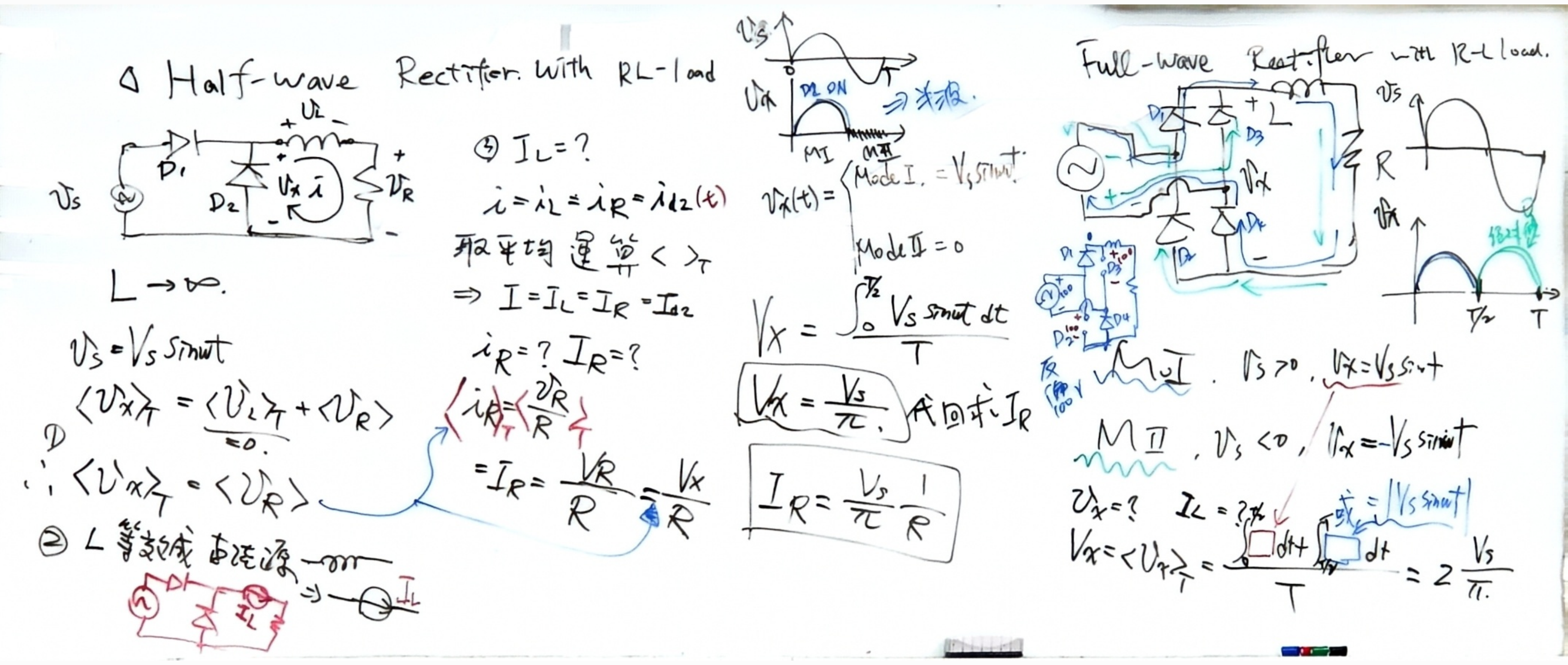


Δ Half-wave Rectifier with RL-load

v_L
D_1
D_2
v_s
v_x
i
v_R
+
−

$L \to \infty$.

$v_s = V_s \sin\omega t$

① $\langle v_x \rangle_T = \underbrace{\langle v_L \rangle_T}_{=0.} + \langle v_R \rangle$

$\therefore \langle v_x \rangle_T = \langle v_R \rangle$

② L 等效成 電流源 $-\!\!\!\text{ooo}\!\!- \Rightarrow$ I_L

③ $I_L = ?$

$i = i_L = i_R = i_{d2}(t)$

取平均運算 $\langle\ \rangle_T$

$\Rightarrow I = I_L = I_R = I_{d2}$

$i_R = ?\ I_R = ?$

$\langle i_R \rangle_T = \langle \frac{v_R}{R} \rangle_T$

$= I_R = \frac{V_R}{R} = \frac{V_x}{R}$

v_s
0
T
v_x
D2 ON
MI
MII

⇒ 濾波.

$$v_x(t) = \begin{cases} \text{Mode I}, = V_s \sin\omega t \\ \text{Mode II} = 0 \end{cases}$$

$$V_x = \frac{\int_0^{T/2} V_s \sin\omega t\, dt}{T}$$

$\boxed{V_x = \frac{V_s}{\pi}.}$ 代回求 I_R

$$\boxed{I_R = \frac{V_s}{\pi}\frac{1}{R}}$$

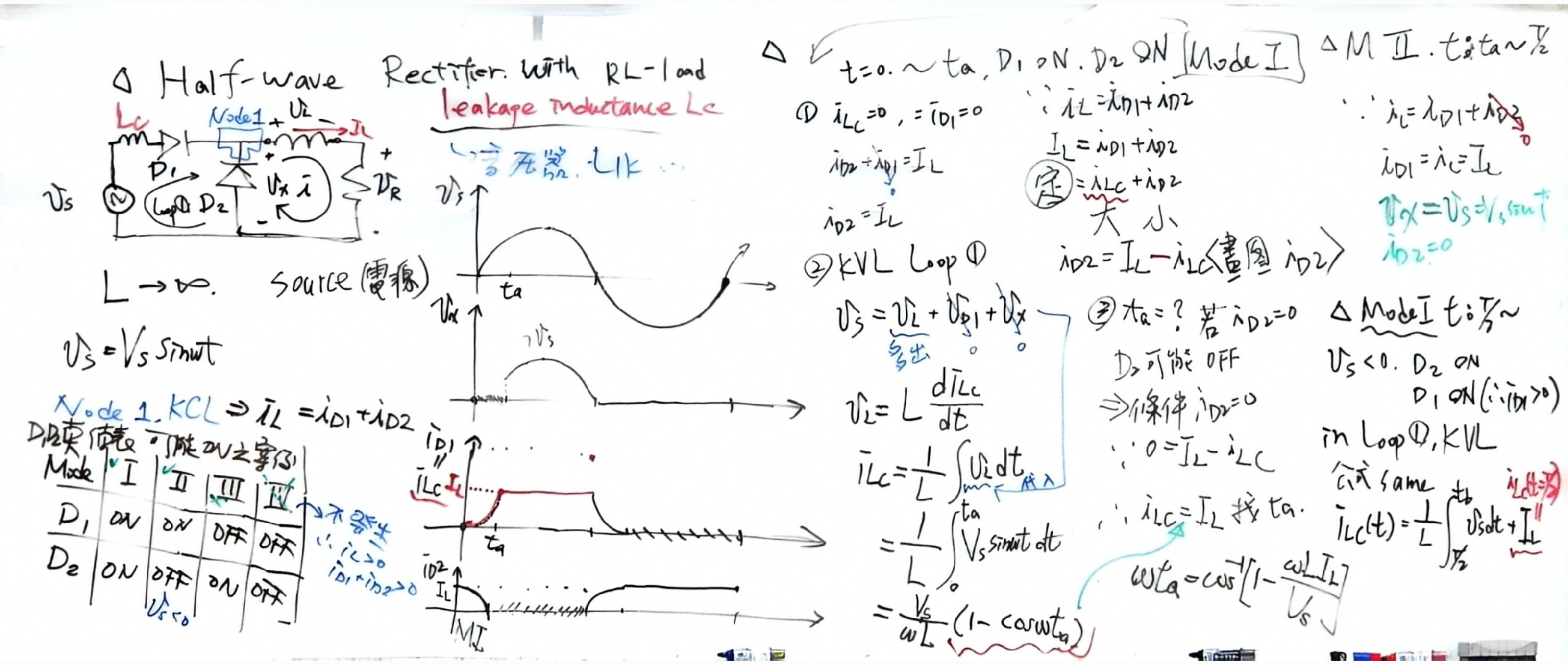
Full-wave Rectifier with R-L load.

MI. $v_s > 0$, $v_x = V_s \sin\omega t$

MII, $v_s < 0$, $v_x = -V_s \sin\omega t$

$v_x = ?$ $I_L = ?$ 或 $= |V_s \sin\omega t|$

$$V_x = \langle v_x \rangle_T = \frac{\int_0 \square\, dt + \int_{T/2} \square\, dt}{T} = 2\frac{V_s}{\pi}.$$



Δ Half-wave Rectifier with RL-load

leakage inductance L_c

↳ 變壓器. L_{lk} ...

L_c
Node1
v_L
I_L
D_1
Loop①
D_2
v_s
v_x
i
v_R
+
−
v_s
t_a
v_x
V_s
i_{D1} = i_{Lc}
I_L
t_a
i_{D2}
I_L
MI

$L \to \infty$. source (電源)

$v_s = V_s \sin\omega t$

Node 1. KCL $\Rightarrow i_L = i_{D1} + i_{D2}$

D1D2 真值表. 可能 ON 之情況

Mode	I	II	III	IV
D_1	ON	ON	OFF	OFF
D_2	ON	OFF $v_s<0$	ON	OFF

→ 不發生 ∵ $i_L > 0$ $i_{D1} + i_{D2} > 0$

Δ $t = 0 \sim t_a$, D_1 ON. D_2 ON [Mode I]

① $i_{Lc} = 0$, $= i_{D1} = 0$

$i_{D2} + i_{D1} = I_L$

$i_{D2} = I_L$

∵ $i_L = i_{D1} + i_{D2}$

$I_L = i_{D1} + i_{D2}$

(定) $= i_{Lc} + i_{D2}$ 大 小

$i_{D2} = I_L - i_{Lc}$ 〈畫圖 i_{D2}〉

② KVL Loop ①

$v_s = \underset{\text{多出}}{v_L} + \underset{0}{v_{D1}} + \underset{0}{v_x}$

$v_L = L\frac{di_{Lc}}{dt}$

$i_{Lc} = \frac{1}{L}\int v_L\, dt$ 代入

$= \frac{1}{L}\int_0^{t_a} V_s \sin\omega t\, dt$

$= \frac{V_s}{\omega L}(1 - \cos\omega t_a)$

③ $t_a = ?$ 若 $i_{D2} = 0$

D_2 可能 OFF

⇒ 條件 $i_{D2} = 0$

∵ $0 = I_L - i_{Lc}$

∴ $i_{Lc} = I_L$ 找 t_a.

$\omega t_a = \cos^{-1}\left[1 - \frac{\omega L I_L}{V_s}\right]$

Δ M II. $t: t_a \sim T/2$

∵ $i_L = i_{D1} + i_{D2} \to 0$

$i_{D1} = i_L = I_L$

$v_x = v_s = V_s \sin\omega t$

$i_{D2} = 0$

Δ Mode I $t: T/2 \sim$

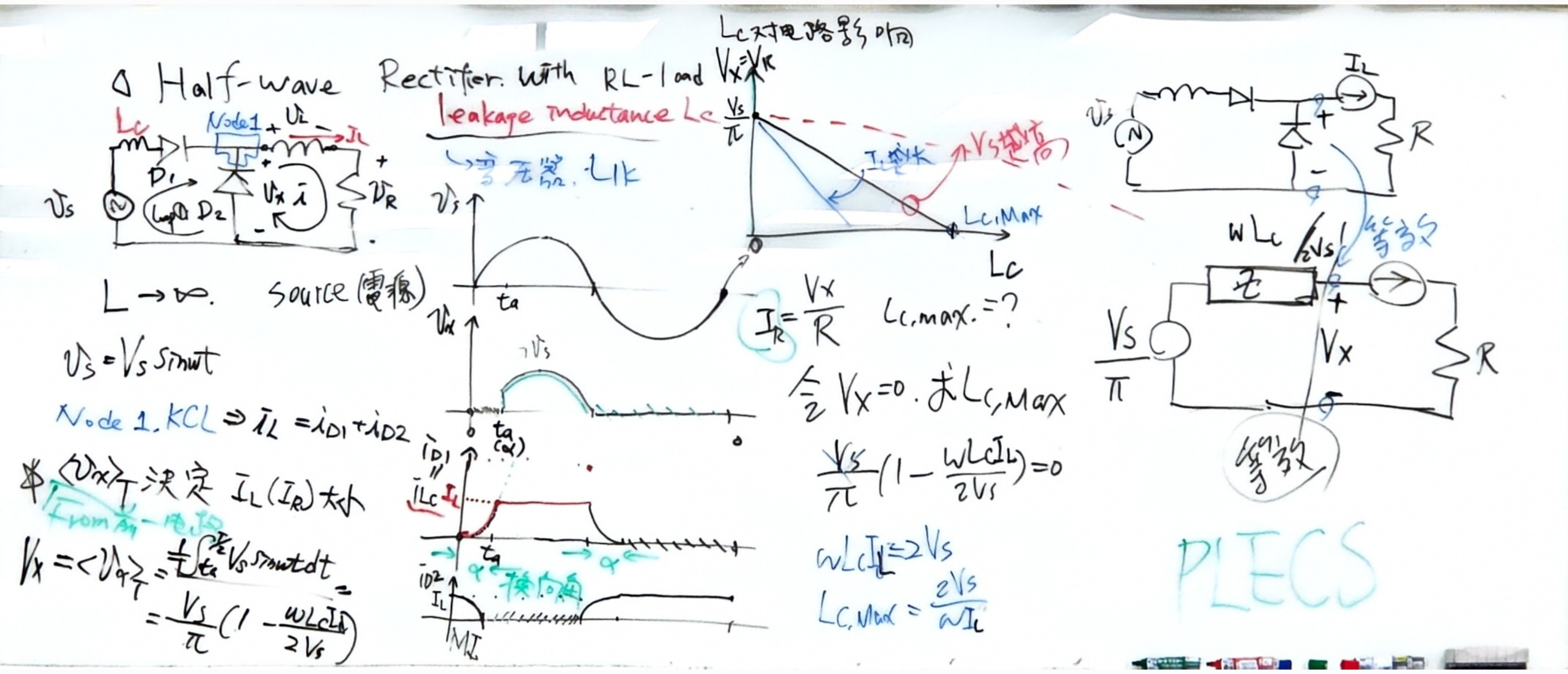
$v_s < 0$. D_2 ON

D_1 ON (∵ $i_{D1} > 0$)

in Loop ①, KVL

公式 same

$i_{Lc}(t) = \frac{1}{L}\int_{T/2}^{t} v_s dt + I_L$ ($i_{Lc}(t = T/2)$)



Δ Half-wave Rectifier with RL-load

leakage inductance L_c

↳ 變壓器. L_{lk}

L_c
Node1
v_L
I_L
D_1
Loop①
D_2
v_s
v_x
i
v_R
+
−
v_s
t_a
v_x
V_s
0
t_a (α)
i_{D1} = i_{Lc}
I_L
t_a
α
α 換向角
i_{D2}
I_L
MI

$L \to \infty$. source (電源)

$v_s = V_s \sin\omega t$

Node 1. KCL $\Rightarrow i_L = i_{D1} + i_{D2}$

* $\langle v_x \rangle_T$ 決定 $I_L(I_R)$ 大小

From 前[illegible]

$V_x = \langle v_x \rangle_T = \frac{1}{T}\int_{t_a}^{T/2} V_s \sin\omega t\, dt =$

$= \frac{V_s}{\pi}\left(1 - \frac{\omega L_c I_L}{2V_s}\right)$

Lc 對電路影響

$V_x = V_R$
$\frac{V_s}{\pi}$
I_L 變大
V_s 越高
$L_{c,Max}$
L_c
0

$I_R = \frac{V_x}{R}$ $L_{c,max} = ?$

令 $V_x = 0$. 求 $L_{c,Max}$

$\frac{V_s}{\pi}\left(1 - \frac{\omega L_c I_L}{2V_s}\right) = 0$

$\omega L_c I_L = 2V_s$

$L_{c,Max} = \frac{2V_s}{\omega I_L}$

PLECS